FINANCIAL ECONOMETRICS

ENSAE

PRACTICAL SESSION n^o 3 - JANUARY 2026

Instructions

This work can be treated alone or in pair. Answers have to be sent in a .pdf file at the email address given by your teaching assistant with the following format: "NAME1,NAME2 - TD3.pdf", **before January 16, at midnight**. Any work sent after this deadline will not be accepted. Particular attention will be given to the quality of the redaction. **Your work should be as concise as possible**. Please join the Excel file with your computations.

Data description

Data for this project are available in the file Data_TD3_2022.xlsx. Spreadsheet "Equity Data" contains the closing prices of Ford (F), Johnson & Johnson (JNJ) and JP Morgan (JPM) stocks. In addition, the series RF contains the daily returns of the risk free asset. Spreadsheet 2 contains daily returns of the five factors of the Fama-French model: the market portfolio (Mkt-RF), "Value" (HML), "Size" (SMB), "Profitability" (RMW) and "Investment" (CMA). The data series range from January 2, 2015, to October 29, 2021.

Part I: Capitalization-weighted portfolio

In addition to the closing prices, Spreadsheet "Equity Data" contains the market capitalization of Ford (F), Johnson & Johnson (JNJ) and JP Morgan (JPM) as of January 2, 2015 in cells G4:I4. A *capitalization-weighted* portfolio (CWP) is constructed such that the weight $w_{i,t}$ of each asset i in the value v_t of the portfolio is its proportion in the global capitalization C_t:

$$w_{i,t} = \frac{v_{i,t}}{v_t} = \frac{n_{i,t}p_{i,t}}{\sum_j n_{j,t}p_{j,t}} = \frac{C_{i,t}}{C_t},$$

where $p_{i,t}$, $n_{i,t}$, $v_{i,t}$ denote respectively the price, the number of shares and the value of asset i in the portfolio, and $C_{i,t}$ is the capitalization of asset i. It is assumed throughout that the number N_i of units of each asset i in the market is fixed: $C_{i,t} = N_i p_{i,t}$.
In the following, all the estimators should be annualized with $T = 252$.

1. Show that a CWP at time $t = 0$ whose composition is fixed (i.e. $n_{i,t} = n_i$) will remain capitalization-weighted.

2. Deduce that for such a portfolio: $\quad v_t = v_0 \sum_i \frac{p_{i,t}}{p_{i,0}} w_{i,0}$.

3. Compute the return and price series of the CWP starting from January 2, 2015 with an initial value of 100.

4. Test the efficiency of this portfolio on October 29, 2021.

5. Estimate the 3-factor model of Fama-French[1] on the portfolio net return series. Propose a test for the 1-factor model (market) against the 3-factor model. Discuss the results of this test.

Part II: Efficient portfolio and investment views

We denote by $\boldsymbol{z}_t = (z_{\mathrm{F},t}, z_{\mathrm{JNJ},t}, z_{\mathrm{JPM},t})'$ the net excess returns of the three stocks with expectation $\tilde{\boldsymbol{\mu}}$ and covariance matrix $\boldsymbol{\Omega}$. In the following, the composition of the portfolios has to be understood in terms of proportions invested in risky assets. For simplicity of notation, the time dependence of the parameters will be omitted. As in the previous section all the estimators should be annualized with $T = 252$.

1. Estimate the efficient portfolio $\overline{\boldsymbol{w}}$ as of October 29, 2021 verifying $\overline{\boldsymbol{w}}'e = 1$.

2. Determine the risk aversion $\overline{\lambda}$ associated to this portfolio, in the presence of a risk free asset.

In the following, we apply the Black-Litterman framework to incorporate views into investment portfolios[2]. Consider an investor Λ who wants to rebalance his/her portfolio on October 29, 2021. To take into account the uncertainty around the value of the expected net excess returns $\tilde{\boldsymbol{\mu}}$, Black and Litterman assume $\tilde{\boldsymbol{\mu}} \sim \mathcal{N}(\boldsymbol{\pi}, \tau\boldsymbol{\Omega})$ where $\boldsymbol{\pi}$ is called the expected equilibrium returns and τ is a positive constant. In addition, we assume $\boldsymbol{z}_t \sim \mathcal{N}(\tilde{\boldsymbol{\mu}}, \boldsymbol{\Omega})$ conditionally on $\tilde{\boldsymbol{\mu}}$. The market portfolio $\boldsymbol{w}^{(m)}$ is the portfolio invested in risky assets such that

$$\boldsymbol{w}^{(m)} = \arg\max_{\boldsymbol{w}} \quad \boldsymbol{w}'\boldsymbol{\pi} - \frac{\overline{\lambda}}{2}\boldsymbol{w}'\boldsymbol{\Omega}\boldsymbol{w}.$$

3. A popular choice for the value of the parameter τ in $\tilde{\boldsymbol{\mu}} \sim \mathcal{N}(\boldsymbol{\pi}, \tau\boldsymbol{\Omega})$ is $\tau = 1/n$ where n is the sample size. Justify this choice.

[1] Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of financial economics*, 33(1), 3-56.

[2] Interested students may refer to Kolm et al. (2021). Black–Litterman and Beyond: The Bayesian Paradigm in Investment Management. *The Journal of Portfolio Management*, 47(5), 91-113. and references therein (available on Pamplemousse) and Meucci, A. (2010). *The Black-Litterman approach: Original model and extensions* available on Google Scholar.

4. Denoting the capitalization-weighted portfolio weights $\boldsymbol{w}^{\mathcal{C}} = (w_{\text{F}}^{\mathcal{C}}, w_{\text{JNJ}}^{\mathcal{C}}, w_{\text{JPM}}^{\mathcal{C}})'$ as of October 29, 2021, compute $\boldsymbol{\pi}$ so that $\boldsymbol{w}^{\mathcal{C}} = \boldsymbol{w}^{(m)}$. Comment this assumption.

Consider now that investor Λ has k subjective *views* on the performance (absolute or relative) of some assets. The Black-Litterman framework allows to deviate from the equilibrium portfolio based on linear *views* on $\boldsymbol{z}_t$. We denote by $\boldsymbol{P}$ the *pick* matrix of size $(k \times 3)$ and by $\boldsymbol{v}$ the so-called view vector of size $(k \times 1)$. For example, investor Λ might think that the Ford stock will outperform the JP Morgan stock by 10% or that the Johnson & Johnson stock will have a negative performance of -5%. In such case,

$$\boldsymbol{P} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \boldsymbol{v} = \begin{pmatrix} 10\% \\ -5\% \end{pmatrix}. \tag{1}$$

To take into account some uncertainty on the views, we consider a random vector

$$\boldsymbol{V} = \boldsymbol{P}\tilde{\boldsymbol{\mu}} + \boldsymbol{\epsilon}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma})$, with $\boldsymbol{\Sigma}$ positive-definite, and is independent from $\tilde{\boldsymbol{\mu}}$.

5. Determine the distribution of $\boldsymbol{V}$, and the distribution of $\boldsymbol{V}$ conditionally on $\tilde{\boldsymbol{\mu}}$.

We also assume that conditionally on $\tilde{\boldsymbol{\mu}}$, the random variables $\boldsymbol{z}_t$ and $\boldsymbol{V}$ are independent. We will incorporate the views of the investor Λ by conditioning the net excess returns on $\boldsymbol{V} = \boldsymbol{v}$.

6. Show that the distribution of $\boldsymbol{z}_t$ conditionally on $\boldsymbol{V} = \boldsymbol{v}$ (i.e. conditionally on investor Λ having the view $\boldsymbol{v}$ associated to the pick matrix $\boldsymbol{P}$) is Gaussian without computing the moments.

One can show that conditionally on $\boldsymbol{V} = \boldsymbol{v}$, the moments of the *posterior* distribution of the net excess return $\boldsymbol{z}_t$ are given by

$$\begin{aligned} \tilde{\boldsymbol{\mu}}^{BL} &= \boldsymbol{\pi} + \tau\boldsymbol{\Omega}\boldsymbol{P}'(\tau\boldsymbol{P}\boldsymbol{\Omega}\boldsymbol{P}' + \boldsymbol{\Sigma})^{-1}(\boldsymbol{v} - \boldsymbol{P}\boldsymbol{\pi}) \\ \boldsymbol{\Omega}^{BL} &= (1+\tau)\boldsymbol{\Omega} - \tau^2\boldsymbol{\Omega}\boldsymbol{P}'(\tau\boldsymbol{P}\boldsymbol{\Omega}\boldsymbol{P}' + \boldsymbol{\Sigma})^{-1}\boldsymbol{P}\boldsymbol{\Omega}. \end{aligned}$$

In the sequel, assume that $\boldsymbol{\Sigma} = \frac{1}{c}\boldsymbol{P}\boldsymbol{\Omega}\boldsymbol{P}'$ with $c > 0$.

7. Implement the views defined in Equation (1) and compute the corresponding $\tilde{\boldsymbol{\mu}}^{BL}$ and $\boldsymbol{\Omega}^{BL}$ for $\tau = 1/n$ and $c = 10$.

8. Compute the optimal portfolio of investor Λ with these views assuming that he/she has risk aversion $\overline{\lambda}$. Compare the results with $c = 1000$ and give an interpretation for parameter c.